Multiple Comparisons: Overview

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Introduction

Testing many hypotheses simultaneously inflates the chance of spurious discoveries. Controlling for this inflation is the goal of multiple-comparison corrections. Two broad families exist: **family-wise error rate (FWER)** methods and **false discovery rate (FDR)** methods. Choosing between them depends on the consequences of a false positive.

Prerequisites

Hypothesis testing, Type I error.

Theory

Suppose m hypotheses are tested and V of them are falsely rejected (true nulls). Let R be total rejections.

- **FWER** $= P(V \geq 1)$. The probability of any false rejection.
- **FDR** $= E[V/\max(R,1)]$. Expected proportion of false positives among rejections.

Bonferroni, Holm, and Hochberg control FWER. Benjamini-Hochberg and Benjamini-Yekutieli control FDR.

When to use which:

- FWER: small number of pre-specified comparisons, confirmatory studies, clinical trial endpoints.
- FDR: exploratory or high-throughput studies with thousands of tests (genomics, imaging), where a few percent false discoveries are acceptable.

Assumptions

Methods assume valid individual p-values (uniform under null). FWER corrections are robust to arbitrary dependence; FDR methods vary (BH needs positive dependence, BY handles arbitrary).

R Implementation

```r
set.seed(2026)
m <- 100
pvals <- runif(m) * ifelse(seq_len(m) <= 10, 0.01, 1)   # 10 small, 90 null

adj_methods <- c("bonferroni", "holm", "hochberg", "BH", "BY")
adj <- sapply(adj_methods, function(m_name)
  p.adjust(pvals, method = m_name))

data.frame(method = adj_methods,
           rejections_at_0.05 = colSums(adj < 0.05))
```

Output & Results

```
      method rejections_at_0.05
```

```
1 bonferroni                  8
2        holm                  8
3    hochberg                  9
4          BH                 10
5          BY                  7
```

FDR (BH) recovers the full signal set; FWER methods are more conservative. BY is tighter than BH because of its dependence-robust adjustment.

Interpretation

In a paper: "Multiple testing across 100 primary comparisons was controlled at FWER 0.05 using the Holm procedure (8 of 100 hypotheses rejected)." Or: "Across 50 000 transcripts, differential expression was declared at Benjamini-Hochberg $q < 0.05$ (2 341 transcripts)."

Practical Tips

- Correct for **all** tests that belong to the same family; a family is defined by the inferential goal, not by software runs.
- Pre-specify the family and the method in the protocol; post-hoc changes to the correction are data-snooping.
- For hierarchically structured hypotheses, gatekeeping procedures maintain FWER at each level.
- Report both raw and adjusted p-values for transparency.
- False discovery proportion (FDP) is the realisation of FDR on a given dataset; FDR is the expectation.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Populations, samples, and the central limit theorem
- Bootstrap and permutation tests
- Maximum likelihood estimation
- One-sample t-test and one-proportion test
- Hypothesis testing, p-values, type I/II errors
- Two-sample and paired t-tests
- Two proportions, chi-square, risk and odds
- Pearson and Spearman correlation
- Non-parametric tests